

IMEN266 Operations Research II

Chapter 8. Queueing Theory

Fall, 2026



INDUSTRIAL AND MANAGEMENT
ENGINEERING, POSTECH

Outline

- 1 Queueing Theory
- 2 Infinite capacity queueing systems
- 3 Finite capacity queueing systems
- 4 Variations and non-Markovian queues
- 5 Time reversibility
- 6 Queueing network

How to study this chapter (2026 format)

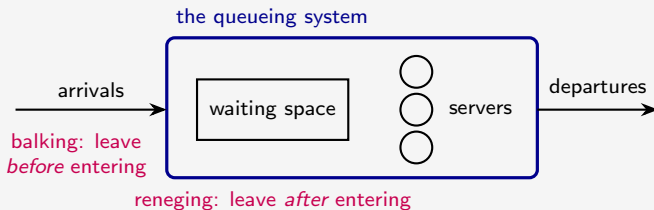
- **Slides** (this file): concepts, the queueing models, and the worked derivations developed in class (p_n , L , W of every model). No transcription needed — definitions, theorems and proofs also live in the Companion.
- **Notebooks** (CP11 Markovian queues · CP12 finite capacity, $M/G/1$, variations · CP13 Jackson networks): in class the INCLASS version (fill the TODOs, `check()` answers instantly); at home the full version with solutions.
- **Companion** (gap-fill $\rightarrow$ full): all definitions and theorems + 8 core proofs with blanks, checkpoints, AI study cards.
- **Homework 5**: A exam-style (no AI first) · B critique an AI solution · C stress-test the formulas by discrete-event simulation · D prompt log.
- Class rhythm (75 min): quiz 5' $\rightarrow$ concepts $\rightarrow$ live notebook $\rightarrow$ wrap-up. Ch.8 = weeks 12–14.
- Ground truth = analytic $\leftrightarrow$ simulation agreement, not AI confidence. Materials: PLMS and GitHub `imen266-2026`.

History

- 1909 A. K. Erlang (1878-1929), a Danish engineer
Modeling the dynamics of a telephone switch.
- 1951 David G. Kendall (1918-2007), English statistician
“Some problems in the theory of queues”,
Journal of the Royal Statistical Society, vol. 31, pp 151-185
first used the term *Queueing System*.
- 1953 *Kendall notation*
- 1970 Donald L. Iglehart, *Limit theorems for queues*
- 1981 Marcel Neuts, *Matrix Analytic Method*
- 1981 S. Halfin, W. Whitt, *Heavy traffic limit*

Kendall notation

The anatomy of a queueing system



To specify: the **arrival process**; the **service-time** law; the **number of servers**; the **capacity** (waiting space + servers); the **service discipline** (FCFS, LCFS, SPTF = shortest processing time first, SRTF = shortest remaining time first, ...).

▷ Kendall's shorthand lists exactly these five items: $A/B/s/K/D$.

Kendall notation

The five fields $A/B/s/K/D$

arrivals / service time / # servers / capacity / discipline

A arrivals	B service	s	K	D	
M Poisson process	M exponential	$1, 2, \dots$	$1, 2, \dots$	FCFS	(default)
G general, GI general independent (= renewal)	G, GI	s	K	LCFS	
D deterministic	D	∞	∞ (default)	SPTF	
E_k Erlang- k , PH phase type	E_k, PH			SRTF	

Two conventions for K

Here K counts the whole *system* (in service + waiting): $M/M/1/K$ holds at most K customers. Some books count the waiting space only.

Kendall notation

Reading examples

$M/M/1$: Poisson arrivals, exponential service, one server, infinite capacity, FCFS (omitted fields take their defaults).

$M/E_3/3/7/LCFS$: Poisson arrivals, Erlang-3 service, 3 servers, at most 7 in the system, last come first served.

$G/G/1 + G$: “+G” is a non-standard tag for a patience-time law — balking and reneging have no standard notation.

What we want from the analysis

Traditional metrics: the number in the system (queue length) and the waiting time — L, L_q, W, W_q , next slide. Since COVID-19 also the infection risk (how many others one overlaps with, for how long), and the psychology of queues: perceived time, information, fairness.

▷ This chapter: FCFS, patient customers; balking/reneging return as CTMCs in CP12 §7 and HW5.

Queueing notation

- λ : average arrival rate
- μ : average service rate
- L : average number of customers in the system
- L_q : average number of customers in the queue
- W : average waiting time of a customer in the system
- W_q : average waiting time of a customer in the queue
- Little's law

$$L = \lambda W$$

$$L_q = \lambda W_q.$$

▷ Companion §1.1 and Proof 3: Little's law $L = \lambda W$ is a renewal reward identity (Ch. 7); $L = L_q + \lambda/\mu$, $W = W_q + 1/\mu$.

Queueing notation

Time averages and customer averages

$X(t)$ = number in the system at time t (waiting + in service), $X_q(t)$ = number waiting; W_n = sojourn time of the n -th customer (wait + service), $W_{q,n}$ = its waiting time:

$$L = \lim_{t \rightarrow \infty} E[X(t)], \quad L_q = \lim_{t \rightarrow \infty} E[X_q(t)] \quad (\text{time averages}),$$

$$W = \lim_{n \rightarrow \infty} E[W_n], \quad W_q = \lim_{n \rightarrow \infty} E[W_{q,n}] \quad (\text{customer averages}).$$

Three identities

A service lasts $1/\mu$ on average, so $W = W_q + \frac{1}{\mu}$; and $L = L_q + \frac{\lambda}{\mu}$, where λ/μ = mean number *in service* = mean number of busy servers. Little's law $L = \lambda W$, $L_q = \lambda W_q$ ties a time average to a customer average (Companion Proof 3).

▷ Plan for every $M/M/\cdot$ model: $p_n \rightarrow L = \sum_n np_n \rightarrow$ the other three.

$M/M/1$ queue

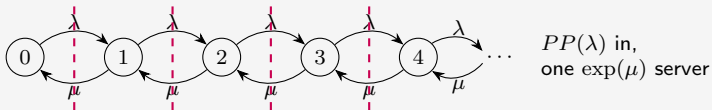
- $X(t) \equiv \#$ of customers in the system at time t .
- $S = \{0, 1, 2, \dots\}$

▷ Companion Proof 1: birth–death stationary distributions from the cut equations;
stability $\rho = \lambda/\mu < 1$.

► Colab notebook: CP11 §1 ($M/M/1$, Little, PASTA) [INCLASS] [full]

M/M/1 queue

Rate diagram and the cut equations



A birth-death CTMC ($\lambda_n = \lambda$, $\mu_n = \mu$). Across each dashed cut the two flows balance:

$$\lambda p_0 = \mu p_1 \Rightarrow p_1 = \frac{\lambda}{\mu} p_0, \quad \lambda p_1 = \mu p_2 \Rightarrow p_2 = \frac{\lambda}{\mu} p_1 = \left(\frac{\lambda}{\mu}\right)^2 p_0,$$

$$\lambda p_2 = \mu p_3 \Rightarrow p_3 = \left(\frac{\lambda}{\mu}\right)^3 p_0, \quad \dots \quad p_n = \left(\frac{\lambda}{\mu}\right)^n p_0, \quad n \geq 0$$

(p_0 from normalisation, next slide).

$M/M/1$ queue

Normalisation and the stability condition

$$p_0 + p_1 + p_2 + \cdots = p_0 \left(1 + \frac{\lambda}{\mu} + \left(\frac{\lambda}{\mu} \right)^2 + \cdots \right) = 1.$$

The geometric series converges iff $\rho := \lambda/\mu < 1$ (**stability condition**: the server is faster than the arrivals). Then

$$p_0 \cdot \frac{1}{1 - \lambda/\mu} = 1 \quad \implies \quad p_0 = 1 - \frac{\lambda}{\mu} = 1 - \rho, \quad p_n = (1 - \rho) \rho^n.$$

Reading ρ

$\rho = \lambda/\mu$ is the **traffic intensity**; $1 - p_0 = \rho$ is the long-run fraction of time the server is busy. For $\rho \geq 1$ no stationary distribution exists — the queue grows without bound (CP11 §1).

M/M/1 queue

L from $p_i = (1 - \rho)\rho^i$: the derivative trick

$$L = \sum_{i=0}^{\infty} i p_i = 0 \cdot p_0 + 1 \cdot p_1 + 2 \cdot p_2 + \cdots = (1 - \rho) \rho (1 + 2\rho + 3\rho^2 + \cdots).$$

For $0 < x < 1$:

$$1 + 2x + 3x^2 + \cdots = \frac{d}{dx} (1 + x + x^2 + \cdots) = \frac{d}{dx} \frac{1}{1 - x} = \frac{1}{(1 - x)^2}, \text{ so}$$

$$L = \frac{(1 - \rho) \rho}{(1 - \rho)^2} = \frac{\rho}{1 - \rho} = \frac{\lambda}{\mu - \lambda}.$$

▷ The same trick, $\sum_i i x^{i-1} = 1/(1 - x)^2$, handles every geometric tail below $(M/M/2, M/M/s)$.

$M/M/1$ queue

W, W_q, L_q by Little's law and the two identities

$$W = \frac{L}{\lambda} = \frac{1}{\mu - \lambda}, \quad W_q = W - \frac{1}{\mu} = \frac{\lambda}{\mu(\mu - \lambda)},$$

$$L_q = \lambda W_q = \frac{\lambda^2}{\mu(\mu - \lambda)} \quad \left(\text{check: } L - L_q = \frac{\lambda}{\mu} = \rho \right).$$

Not only the mean: in steady state the sojourn time of a customer has the $\exp(\mu - \lambda)$ distribution (Companion Proof 4: a geometric number of exponential phases).

The blow-up as $\rho \uparrow 1$

$L = \rho/(1 - \rho)$: $\rho = 0.5, 0.9, 0.99 \Rightarrow L = 1, 9, 99$. Near $\rho = 1$ a small increase in load multiplies the queue — and a simulation needs a run length growing like $1/(1 - \rho)^2$ to measure it (CP11 §1).

$M/M/2$ queue

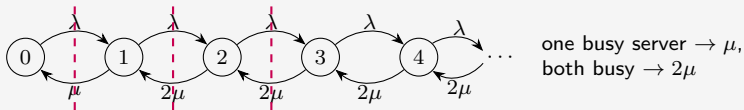
- $X(t) \equiv \#$ of customers in the system at time t .
- $S = \{0, 1, 2, \dots\}$

▷ Companion Proof 2: $M/M/s$ with $s = 2$; Proof 2(c) compares two servers with one server of double speed.

► Colab notebook: CP11 §2 (two servers vs. one fast) [INCLASS] [full]

M/M/2 queue

Rate diagram: the death rate saturates at 2μ



$$\lambda p_0 = \mu p_1 \Rightarrow p_1 = \frac{\lambda}{\mu} p_0, \quad \lambda p_1 = 2\mu p_2 \Rightarrow p_2 = \frac{\lambda}{2\mu} p_1 = \frac{\lambda}{2\mu} \cdot \frac{\lambda}{\mu} p_0,$$

$$\lambda p_2 = 2\mu p_3 \Rightarrow p_3 = \frac{\lambda}{2\mu} p_2 = \left(\frac{\lambda}{2\mu}\right)^2 \frac{\lambda}{\mu} p_0, \quad \dots$$

$$p_i = \left(\frac{\lambda}{2\mu}\right)^{i-1} \frac{\lambda}{\mu} p_0 \quad (i \geq 1)$$

▷ Only the first ratio is λ/μ ; from state 1 on the ratio is $\lambda/(2\mu)$ — this is what stability will depend on.

M/M/2 queue

Normalisation and stability

$$\begin{aligned} 1 &= p_0 \left(1 + \frac{\lambda}{\mu} + \frac{\lambda}{\mu} \frac{\lambda}{2\mu} + \frac{\lambda}{\mu} \left(\frac{\lambda}{2\mu} \right)^2 + \dots \right) \\ &= p_0 \left(1 + \frac{\lambda}{\mu} \left(1 + \frac{\lambda}{2\mu} + \left(\frac{\lambda}{2\mu} \right)^2 + \dots \right) \right). \end{aligned}$$

If $\lambda/(2\mu) < 1$ (stability: two servers out-pace the arrivals), the inner series is $1/(1 - \lambda/(2\mu))$ and

$$p_0 \left(1 + \frac{\lambda}{\mu} \cdot \frac{1}{1 - \lambda/(2\mu)} \right) = 1 \quad \implies \quad p_0 = \frac{2\mu - \lambda}{2\mu + \lambda}.$$

Stationary distribution

$$p_i = \frac{\lambda}{\mu} \left(\frac{\lambda}{2\mu} \right)^{i-1} \frac{2\mu - \lambda}{2\mu + \lambda}, \quad i = 1, 2, 3, \dots$$

M/M/2 queue

L , and the rest as an exercise

$$\begin{aligned}
 L = \sum_{i \geq 0} i p_i &= \frac{\lambda}{\mu} \cdot \frac{2\mu - \lambda}{2\mu + \lambda} \left(1 + 2 \frac{\lambda}{2\mu} + 3 \left(\frac{\lambda}{2\mu} \right)^2 + \dots \right) \\
 &= \frac{\lambda}{\mu} \cdot \frac{2\mu - \lambda}{2\mu + \lambda} \cdot \frac{1}{(1 - \lambda/(2\mu))^2} = \frac{\lambda}{\mu} \cdot \frac{2\mu - \lambda}{2\mu + \lambda} \cdot \frac{4\mu^2}{(2\mu - \lambda)^2} \\
 &= \frac{4\lambda\mu}{4\mu^2 - \lambda^2}.
 \end{aligned}$$

$$\begin{aligned}
 W, W_q, L_q \text{ (exercise): } W &= \frac{L}{\lambda} = \frac{4\mu}{4\mu^2 - \lambda^2}, \quad L_q = L - \frac{\lambda}{\mu} = \frac{\lambda^3}{\mu(4\mu^2 - \lambda^2)}, \\
 W_q &= \frac{L_q}{\lambda} = \frac{\lambda^2}{\mu(4\mu^2 - \lambda^2)}.
 \end{aligned}$$

▷ CP11 §2 checks these by simulation and compares two servers of rate μ with one server of rate 2μ .

$M/M/s$ queue

- $X(t) \equiv \#$ of customers in the system at time t .
- $S = \{0, 1, 2, \dots\}$

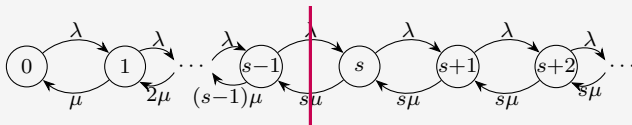
▷ Companion Proof 2: p_0 , L_q and the Erlang C probability of waiting

$$P(W_q > 0) = p_s / (1 - \rho).$$

► Colab notebook: CP11 §3 ($M/M/s$, Erlang C) [INCLASS] [full]

M/M/s queue

Rate diagram: death rate $\min(n, s) \mu$ — two regimes



Below s (left of the bar; the ratios $\lambda/(n\mu)$ shrink):

$$p_1 = \frac{\lambda}{\mu} p_0, \quad p_2 = \frac{\lambda}{2\mu} p_1 = \frac{1}{2!} \left(\frac{\lambda}{\mu}\right)^2 p_0, \quad p_3 = \frac{1}{3 \cdot 2} \left(\frac{\lambda}{\mu}\right)^3 p_0, \quad \dots \quad p_s = \frac{1}{s!} \left(\frac{\lambda}{\mu}\right)^s p_0$$

From s on (all servers busy, constant ratio $\lambda/(s\mu)$):

$$p_{s+1} = \frac{\lambda}{s\mu} p_s, \quad p_{s+2} = \frac{\lambda}{s\mu} p_{s+1} = \left(\frac{\lambda}{s\mu}\right)^2 p_s, \quad \dots \quad p_{s+i} = \left(\frac{\lambda}{s\mu}\right)^i p_s$$

M/M/s queue

Normalisation: a finite sum plus a geometric tail

$$\begin{aligned}
 1 = \sum_i p_i &= p_0 + \underbrace{\frac{\lambda}{\mu} p_0 + \cdots + \frac{1}{s!} \left(\frac{\lambda}{\mu}\right)^s p_0}_{\text{states } 0, \dots, s} + \underbrace{\frac{\lambda}{s\mu} p_s + \left(\frac{\lambda}{s\mu}\right)^2 p_s + \cdots}_{\text{states } s+1, s+2, \dots} \\
 &= p_0 \left(\sum_{i=0}^s \frac{1}{i!} \left(\frac{\lambda}{\mu}\right)^i + \frac{1}{s!} \left(\frac{\lambda}{\mu}\right)^s \left[\frac{\lambda}{s\mu} + \left(\frac{\lambda}{s\mu}\right)^2 + \cdots \right] \right).
 \end{aligned}$$

If $\lambda/(s\mu) < 1$ (stability: s servers out-pace the arrivals) the bracket is $\frac{\lambda/(s\mu)}{1-\lambda/(s\mu)}$ and

$$p_0 \underbrace{\left(\sum_{i=0}^s \frac{1}{i!} \left(\frac{\lambda}{\mu}\right)^i + \frac{1}{s!} \left(\frac{\lambda}{\mu}\right)^s \frac{\lambda/(s\mu)}{1-\lambda/(s\mu)} \right)}_{=:A} = 1 \implies p_0 = \frac{1}{A}.$$

$M/M/s$ queue

L_q directly: only the states above s have a queue

$$\begin{aligned}
 L_q &= 0 \cdot p_0 + \cdots + 0 \cdot p_s + 1 \cdot p_{s+1} + 2 \cdot p_{s+2} + \cdots = \sum_{i=1}^{\infty} i p_{s+i} \\
 &= \sum_{i=1}^{\infty} i p_0 \frac{1}{s!} \left(\frac{\lambda}{\mu}\right)^s \left(\frac{\lambda}{s\mu}\right)^i = p_0 \frac{1}{s!} \left(\frac{\lambda}{\mu}\right)^s \frac{\lambda}{s\mu} \sum_{i \geq 1} i \left(\frac{\lambda}{s\mu}\right)^{i-1} \\
 &= \frac{1}{s!} \left(\frac{\lambda}{\mu}\right)^s \frac{\lambda}{s\mu} \left(\frac{s\mu}{s\mu - \lambda}\right)^2 p_0
 \end{aligned}$$

(the last sum is $1/(1 - \lambda/(s\mu))^2$ by the derivative trick).

Then L , W , W_q — and the probability of waiting

$$L = L_q + \lambda/\mu, \quad W_q = L_q/\lambda, \quad W = W_q + 1/\mu;$$

$$P(\text{wait}) = \sum_{i \geq s} p_i = \frac{p_s}{1 - \lambda/(s\mu)} \quad (\text{Erlang C; CP11 §3}).$$

$M/M/\infty$ queue

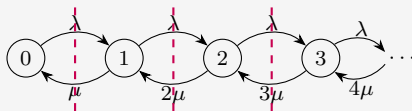
- $X(t) \equiv \#$ of customers in the system at time t .
- $S = \{0, 1, 2, \dots\}$

▷ Companion Proof 1(c): the Poisson stationary law; Proof 6(c) truncates it to $M/M/K/K$.

► Colab notebook: CP11 §4 ($M/M/\infty$) [INCLASS] [full]

$M/M/\infty$ queue

Rate diagram: every customer has its own server, death rate $n\mu$



self-service: nobody waits;
 n customers = n running clocks

$$p_1 = \frac{\lambda}{\mu} p_0, \quad p_2 = \frac{\lambda}{2\mu} p_1 = \frac{1}{2!} \left(\frac{\lambda}{\mu}\right)^2 p_0, \quad p_3 = \frac{\lambda}{3\mu} p_2 = \frac{1}{3!} \left(\frac{\lambda}{\mu}\right)^3 p_0, \quad \dots$$

$$p_i = \frac{1}{i!} \left(\frac{\lambda}{\mu}\right)^i p_0, \quad i = 0, 1, 2, \dots$$

— the first regime of $M/M/s$, which here never ends: no stability condition to check.

$M/M/\infty$ queue

Normalisation gives a Poisson distribution

$$\sum_i p_i = p_0 \left[1 + \frac{\lambda}{\mu} + \frac{1}{2!} \left(\frac{\lambda}{\mu} \right)^2 + \frac{1}{3!} \left(\frac{\lambda}{\mu} \right)^3 + \cdots \right] = 1,$$

and $\sum_{i \geq 0} (\lambda/\mu)^i / i! = e^{\lambda/\mu}$, so

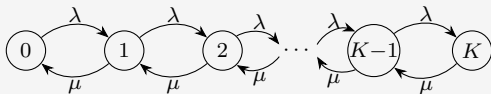
$$p_0 = e^{-\lambda/\mu}, \quad p_i = \frac{1}{i!} \left(\frac{\lambda}{\mu} \right)^i e^{-\lambda/\mu} = \frac{e^{-\lambda/\mu} (\lambda/\mu)^i}{i!}$$

— the Poisson pmf with mean λ/μ .

Performance measures

$$L = \sum_i i p_i = \frac{\lambda}{\mu}, \quad W = \frac{L}{\lambda} = \frac{1}{\mu} \text{ (the service time)}, \quad L_q = W_q = 0.$$

▷ The Poisson law survives for any service-time distribution with mean $1/\mu$ ($M/G/\infty$) — an insensitivity met again in $M/G/K/K$.

$M/M/1/K$ queueFinite capacity K : the chain stops at K 

$S = \{0, 1, \dots, K\}$; an arrival that finds K customers is **lost**

The same cut equations as $M/M/1$, but only K of them — a finite sum, so *no stability condition*:

$$p_1 = \frac{\lambda}{\mu} p_0, \quad p_2 = \left(\frac{\lambda}{\mu}\right)^2 p_0, \quad \dots, \quad p_K = \left(\frac{\lambda}{\mu}\right)^K p_0; \quad p_0 \sum_{i=0}^K \left(\frac{\lambda}{\mu}\right)^i = 1$$

$$\Rightarrow p_0 = \frac{1 - \lambda/\mu}{1 - (\lambda/\mu)^{K+1}}, \quad p_i = \frac{(1 - \lambda/\mu) (\lambda/\mu)^i}{1 - (\lambda/\mu)^{K+1}}, \quad i = 0, \dots, K$$

$$(\lambda = \mu: p_i = 1/(K+1)).$$

► Colab notebook: CP12 §1 ($M/M/1/K$) [INCLASS] [full]

$M/M/1/K$ queue

The number that matters, and Little's law with the *effective* arrival rate

$$p_K = \frac{(1 - \lambda/\mu)(\lambda/\mu)^K}{1 - (\lambda/\mu)^{K+1}} = \text{probability that the system is full} =$$

(PASTA) fraction of arrivals that are **dropped**.

$L = \sum_{i=0}^K i p_i$ (finite sum). $W = L/\lambda$? **No**: a fraction p_K of the arrivals never enters and has no sojourn time. The customers who enter arrive at the **actual (effective) rate** $\lambda(1 - p_K)$, so

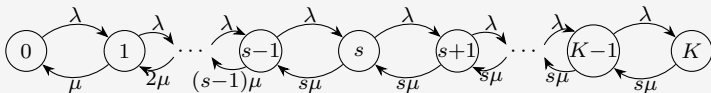
$$W = \frac{L}{\lambda(1 - p_K)}, \quad W_q = W - \frac{1}{\mu}, \quad L_q = \lambda(1 - p_K) W_q.$$

Example (CP12 §1)

$\lambda = \mu$, $K = 4$: $p_i = \frac{1}{5}$, $p_K = 0.2$, $L = 2$, $\lambda(1 - p_K) = 0.8\lambda$, $W = 2.5/\lambda$ — the naive $L/\lambda = 2/\lambda$ is 20% too short.

$M/M/s/K$ queue

$s \leq K$: the $M/M/s$ diagram cut off at K



The cut equations are those of $M/M/s$ (slides 17–19) up to K — a finite sum, no stability condition:

$$p_i = \begin{cases} \frac{1}{i!} \left(\frac{\lambda}{\mu}\right)^i p_0, & 0 \leq i \leq s, \\ \frac{1}{s!} \left(\frac{\lambda}{\mu}\right)^s \left(\frac{\lambda}{s\mu}\right)^{i-s} p_0, & s \leq i \leq K, \end{cases}$$

$$p_0 = \left(\sum_{i=0}^K \frac{p_i}{p_0} \right)^{-1}.$$

► Colab notebook: CP12 §2 ($M/M/s/K$, Example $M/M/5/5$) [INCLASS] [full]

$M/M/s/K$ queue

Performance measures — again with the effective arrival rate

$$L = \sum_{i=0}^K i p_i, \quad W = \frac{L}{\lambda(1 - p_K)}, \quad W_q = W - \frac{1}{\mu}, \quad L_q = \lambda(1 - p_K) W_q$$

($L_q = \sum_{i>s} (i - s) p_i$ directly). p_K = fraction of customers lost;
 $\lambda(1 - p_K)$ = throughput = rate of service completions.

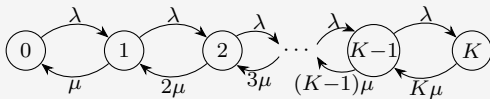
Check

$L - L_q = \lambda(1 - p_K)/\mu$ = mean number of busy servers: every served customer occupies a server for $1/\mu$ on average (Little's law for the servers).

▷ Recipe for every finite-capacity queue: p_K first, then $\lambda(1 - p_K)$, then Little.

$M/M/K/K$ queue

The loss system of A. K. Erlang: K servers, no waiting room



$M/M/\infty$ cut off at K : a call that finds all K lines busy is lost

$$p_1 = \frac{\lambda}{\mu} p_0, \quad p_2 = \frac{1}{2!} \left(\frac{\lambda}{\mu} \right)^2 p_0, \quad \dots, \quad p_i = \frac{1}{i!} \left(\frac{\lambda}{\mu} \right)^i p_0, \quad \dots, \quad p_K = \frac{1}{K!} \left(\frac{\lambda}{\mu} \right)^K p_0;$$

$$p_0 \sum_{i=0}^K \frac{1}{i!} \left(\frac{\lambda}{\mu} \right)^i = 1 \quad \Rightarrow \quad p_0 = \left(\sum_{i=0}^K \frac{1}{i!} \left(\frac{\lambda}{\mu} \right)^i \right)^{-1}.$$

► Colab notebook: CP12 §3 (Erlang loss) [INCLASS] [full]

$M/M/K/K$ queue

Stationary distribution and the Erlang loss formula

$$p_i = \frac{\frac{1}{i!}(\lambda/\mu)^i}{\sum_{j=0}^K \frac{1}{j!}(\lambda/\mu)^j}, \quad i = 0, \dots, K \quad \text{— works for } M/G/K/K \text{ too.}$$

$$p_K = \frac{\frac{1}{K!}(\lambda/\mu)^K}{\sum_{j=0}^K \frac{1}{j!}(\lambda/\mu)^j}$$

Erlang loss formula (Erlang B)

= fraction of calls that find all K lines busy and are lost; only the mean call length $1/\mu$ enters.

Performance

No waiting: $W = 1/\mu$, $W_q = L_q = 0$, $L = \lambda(1 - p_K)/\mu$ = mean number of busy lines. The insensitivity (same p_i for any service law with mean $1/\mu$) is checked by simulation in CP12 §3.

Example $M/M/5/5$ queue

- $\lambda = 5$ per hour
- $1/\mu = 12$ minutes

1. What fraction of customers is rejected?
2. $L = ?$
3. $W = ?$

▷ Companion Proof 5: finite capacity — p_K is the loss probability; Little's law needs the effective arrival rate $\lambda(1 - p_K)$.

► Colab notebook: CP12 §2 (Example $M/M/5/5$) [INCLASS] [full]

Example $M/M/5/5$ queue

Solution: $\lambda = 5/h$, $1/\mu = 12 \text{ min} \Rightarrow \mu = 5/h$, so $\lambda/\mu = 1$

- ❶ Fraction rejected = p_5 (Erlang B with $K = 5$, $\lambda/\mu = 1$):

$$p_5 = \frac{\frac{1}{5!} 1^5}{\sum_{i=0}^5 \frac{1}{i!} 1^i} = \frac{1/120}{1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120}} = \frac{1/120}{326/120} = \frac{1}{326}$$

$\approx 0.31\%$ of the customers.

- ❷ $L = 0 \cdot p_0 + 1 \cdot p_1 + \cdots + 5 \cdot p_5 = \frac{325}{326} \approx 0.997$ (shortcut:

$$L = \lambda(1 - p_K)/\mu = 1 - \frac{1}{326}).$$

- ❸ $W = \frac{L}{\lambda(1 - p_K)} = \frac{325/326}{5(1 - \frac{1}{326})} = \frac{325/326}{5 \cdot 325/326} = \frac{1}{5} \text{ h} = 12 \text{ min}$
 — nobody waits in a loss system, so $W = 1/\mu$.

▷ Dividing L by λ instead of $\lambda(1 - p_K)$ gives 11.96 min, a sojourn shorter than a service — the sign of a forgotten p_K .

Bulk arrival: airport

- $M^{[X]}/M/1$ queue
- Poisson arrivals: $PP(\lambda)$
- Each arrival brings X customers.

$$P(X = i) = q_i, \text{ for } i = 1, 2, 3, \dots$$

- Service is individual.
- Results

$$L = \frac{\rho + \left(\frac{\lambda}{\mu} E[X^2] \right)}{2(1 - \rho)}, \text{ where } \rho = \frac{\lambda}{\mu} E[X].$$

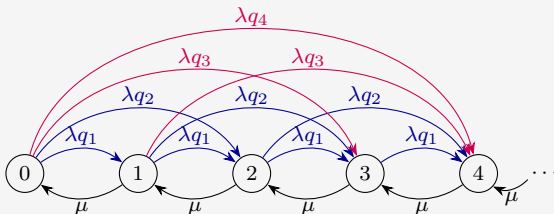
▷ Companion Proof 7(c): the $M^{[X]}/M/1$ formula; $X \equiv 1$ gives back $M/M/1$.

Numbers of the airport example: derivation frame.

► Colab notebook: CP12 §6 (bulk arrivals) [INCLASS] [full]

Bulk arrival: airport

Rate diagram of $M^{[X]}/M/1$ — not a birth–death chain



A batch of size i (probability q_i) moves the state up by i : arcs from n to $n+1, n+2, \dots$ with rates $\lambda q_1, \lambda q_2, \dots$; departures are single, rate μ . A cut now crosses several arcs, so the two-term recursion is gone — the balance equations are solved with a generating function (Companion Proof 7(c)); the slide quotes the result. $X \equiv 1$ gives back $M/M/1$.

▷ **Stability:** $\rho = \lambda E[X]/\mu < 1$ — customers, not batches, must be served faster than they arrive.

Bulk arrival: airport

Numbers: batches of 1, 2, 3, 4 w.p. 0.4, 0.3, 0.2, 0.1; $\lambda = 2$ batches/min, $\mu = 5$ /min

$$E[X] = 0.4 + 0.6 + 0.6 + 0.4 = 2, \quad E[X^2] = 0.4 + 1.2 + 1.8 + 1.6 = 5, \\ \rho = \lambda E[X] / \mu = 2 \cdot 2 / 5 = 0.8;$$

$$L = \frac{\rho + \frac{\lambda}{\mu} E[X^2]}{2(1 - \rho)} = \frac{0.8 + \frac{2}{5} \cdot 5}{2 \cdot 0.2} = 7 \text{ customers}, \quad L_q = L - \rho = 6.2,$$

$$W = \frac{L}{\lambda E[X]} = \frac{7}{4} = 1.75 \text{ min}, \quad W_q = \frac{L_q}{\lambda E[X]} = \frac{6.2}{4} = 1.55 \text{ min}.$$

Two things to notice

Little's law uses the *customer* arrival rate $\lambda E[X] = 4$ /min, not the batch rate. Single arrivals at the same customer rate ($M/M/1$, $\rho = 0.8$) would give $L = 4$: batching almost doubles the congestion at the same utilisation (CP12 §6).

$M/G/1$ queue

- Poisson arrivals: $PP(\lambda)$.
- General service times following $G(\cdot)$ with mean $1/\mu$ and variance σ^2 .
- Results

$$L_q = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1 - \rho)}.$$

▷ Companion Proof 7: Pollaczek–Khinchine from the mean residual service time (Ch. 7 inspection paradox) + PASTA + Little.

► Colab notebook: CP12 §5 ($M/G/1$, P-K formula) [INCLASS] [full]

▷ $\rho = \lambda/\mu$. The result comes from the embedded DTMC at departure epochs (the number left behind by the n -th departure is Markov); $G/M/1$ is treated the same way at arrival epochs.

M/G/1 queue

Where the formula comes from: a mean-value argument

An arrival waits for (i) the *residual* service time R of the customer in service, if any, and (ii) the full services of the L_q customers ahead of it:

$$W_q = E[R] + L_q \cdot \frac{1}{\mu}.$$

By PASTA the arrival sees time averages; by Ch. 7 (inspection paradox) the residual service averages $E[S^2]/(2E[S])$ *while the server is busy*, which is a fraction ρ of the time:

$$E[R] = \rho \frac{E[S^2]}{2E[S]} = \frac{\lambda E[S^2]}{2}, \quad E[S^2] = \sigma^2 + \frac{1}{\mu^2}.$$

With Little, $L_q = \lambda W_q$: $W_q = \frac{\lambda E[S^2]}{2} + \rho W_q$, so

$$W_q = \frac{\lambda E[S^2]}{2(1-\rho)}, \quad L_q = \lambda W_q = \frac{\lambda^2 \sigma^2 + \rho^2}{2(1-\rho)}.$$

$M/G/1$ queue

Reading the Pollaczek–Khinchine formula

Only the mean and the **variance** of the service time matter: $\sigma^2 = 1/\mu^2$ (exponential) gives back $M/M/1$; $\sigma^2 = 0$ ($M/D/1$) halves L_q ; a long-tailed service law can make L_q huge at the same ρ .

Numbers: a web server

$\lambda = 20$ requests/min $= \frac{1}{3}$ /s, $E[S] = 2$ s, $\sigma = 5$ s (Pareto-like file sizes):
 $\rho = \frac{2}{3}$,

$$L_q = \frac{(\frac{1}{3})^2 \cdot 25 + (\frac{2}{3})^2}{2 \cdot \frac{1}{3}} = 4.83, \quad W_q = 14.5 \text{ s}, \quad W = 16.5 \text{ s}, \quad L = 5.5;$$

with exponential service of the same mean, W_q would be 4 s.

▷ CP12 §5 simulates $M/D/1$, $M/M/1$ and a heavy-tailed $M/G/1$ at the same ρ against the formula.

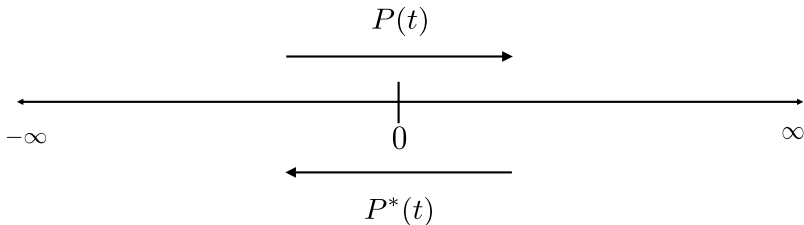
$G/G/1$ queue

- $T \equiv$ random variable of inter-arrival time with $E[T] = 1/\lambda$.
- $S \equiv$ random variable of service time with $E[S] = 1/\mu$.
- Results: no closed form — approximations only. See the handout *The $G/G/1$ and $G/G/m$ queueing systems* on PLMS (summarised in Companion Appendix A).

▷ Companion Appendix A: the $G/G/1$ and $G/G/m$ approximations of the PLMS handout (squared coefficients of variation C_a^2, C_s^2).

► Colab notebook: CP12 §8 ($G/G/1$ approximations) [INCLASS] [full]

Reversed process



▷ Companion Proof 6(a): the reversed chain has rates $q_{ij}^* = p_j q_{ji} / p_i$ — the derivation frame follows.

Reversed process

Transition probabilities of the process run backwards (stationary CTMC)

Let X be stationary, $P(X(t) = i) = p_i$ for all t . Looking *backwards* from time t :

$$\begin{aligned} P_{ij}^*(t) &= P(X(0) = j \mid X(t) = i) = \frac{P(X(0) = j, X(t) = i)}{P(X(t) = i)} \\ &= \frac{P(X(t) = i \mid X(0) = j) P(X(0) = j)}{P(X(t) = i)} = \frac{P_{ji}(t) p_j}{p_i}. \end{aligned}$$

Hence $p_i P_{ij}^*(t) = p_j P_{ji}(t)$. Divide by t , let $t \downarrow 0$ ($P_{ij}(t)/t \rightarrow q_{ij}$ for $i \neq j$):

$$\lim_{t \downarrow 0} \frac{p_i P_{ij}^*(t)}{t} = \lim_{t \downarrow 0} \frac{p_j P_{ji}(t)}{t} \implies p_i q_{ij}^* = p_j q_{ji}.$$

▷ The reversed process is a CTMC with rates $q_{ij}^* = p_j q_{ji}/p_i$;
time-reversible means $q_{ij}^* = q_{ij}$.

Reversible process

Definition

A CTMC is said to be *time-reversible* if

$$p_i q_{ij} = p_j q_{ji}, \text{ i.e., } q_{ij}^* = q_{ij}.$$

▷ Companion Proof 6(b): birth–death chains are reversible (cut equations = detailed balance); Burke's theorem.

Reversible process

Birth–death chains are time-reversible

$p_i q_{ij} = p_j q_{ji}$ are the **arc (detailed) balance** equations. In a birth–death chain the only arcs are $n \leftrightarrow n+1$, and the cut equations $\lambda_n p_n = \mu_{n+1} p_{n+1}$ are these equations: in steady state $M/M/1$, $M/M/s$, $M/M/\infty$ and their finite-capacity versions are all time-reversible.

Burke's theorem



Run the stationary $M/M/s$ backwards: its departures become the arrivals of the reversed process — which is the same process, so they form a $PP(\lambda)$: the departure process of a stationary $M/M/s$ is Poisson(λ), and the departures before t are independent of $X(t)$.

Truncation

- A : Time-reversible CTMC on $\mathcal{S}$
- B : Time-reversible CTMC on $\mathcal{S}'$
- $\mathcal{S}' \subset \mathcal{S}$, truncation of $\mathcal{S}$
- p^A : steady-state distribution of A
- p^B : steady-state distribution of B
- We can obtain p^B from p^A using the following:

$$p_j^B = \frac{p_j^A}{\sum_{i \in \mathcal{S}'} p_i^A}, \text{ for } j \in \mathcal{S}'.$$

▷ Companion Proof 6(c) (final-style): proof of the truncation theorem; $M/M/2/K$ from $M/M/2$, $M/M/K/K$ from $M/M/\infty$.

► Colab notebook: CP12 §4 ($M/M/2/K$ by truncation) [INCLASS] [full]

Truncation

Why truncation works (proof)

On $\mathcal{S}'$ keep every rate q_{ij} with $i, j \in \mathcal{S}'$ and delete the arcs that leave $\mathcal{S}'$.
Candidate: $p_j^B = p_j^A / \sum_{i \in \mathcal{S}'} p_i^A$.

For $i, j \in \mathcal{S}'$ reversibility of A gives $p_i^A q_{ij} = p_j^A q_{ji}$; dividing by the constant $\sum_{\mathcal{S}'} p_i^A$:

$$p_i^B q_{ij} = p_j^B q_{ji} \quad \text{— the detailed balance equations of } B.$$

Summing over i : $\sum_i p_i^B q_{ij} = p_j^B \sum_i q_{ji}$, i.e. the global balance $p^B Q_B = 0$. So p^B is the stationary distribution of B , and B is reversible too. $\square$

▷ Nothing is solved: the weights of A are renormalised on $\mathcal{S}'$.

Reversibility of A is essential — for a non-reversible chain the truncated weights are wrong.

Truncation

Final-style: $M/M/2/K$ from $M/M/2$ — without solving $pQ = 0$

$M/M/2$ weights (slide 14): $p_0 \propto 1$, $p_j \propto \frac{\lambda}{\mu} \left(\frac{\lambda}{2\mu}\right)^{j-1}$ for $j \geq 1$. Truncate to $\mathcal{S}' = \{0, 1, \dots, K\}$:

$$p_j^{(K)} = \frac{\frac{\lambda}{\mu} \left(\frac{\lambda}{2\mu}\right)^{j-1}}{1 + \frac{\lambda}{\mu} \sum_{i=1}^K \left(\frac{\lambda}{2\mu}\right)^{i-1}} \quad (1 \leq j \leq K), \quad p_0^{(K)} = \frac{1}{1 + \frac{\lambda}{\mu} \cdot \frac{1 - (\lambda/2\mu)^K}{1 - \lambda/2\mu}}$$

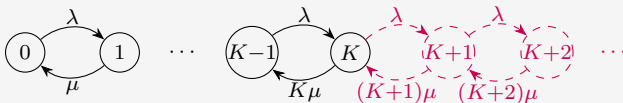
($\lambda \neq 2\mu$). Only the *ratios* of the $M/M/2$ weights enter, so the stability condition $\lambda < 2\mu$ is not needed.

$\lambda = \mu$, $K = 4$: $p^{(4)} = (0.348, 0.348, 0.174, 0.087, 0.043)$, loss $p_K = 4.3\%$.

▷ A 2025 final-exam question. CP12 §4 checks the answer against the direct solution of $pQ = 0$.

$M/M/\infty$ and $M/M/K/K$ queues

$M/M/K/K$ is the truncation of $M/M/\infty$ to $\mathcal{S}' = \{0, 1, \dots, K\}$



$M/M/\infty$ on $\mathcal{S} = \{0, 1, 2, \dots\}$: $p_j^\infty = \frac{(\lambda/\mu)^j e^{-\lambda/\mu}}{j!}$, reversible

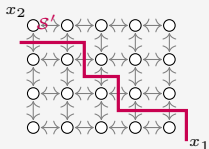
(birth–death). Deleting the dashed part leaves exactly the $M/M/K/K$ diagram, so

$$p_j^K = \frac{p_j^\infty}{\sum_{i=0}^K p_i^\infty} = \frac{(\lambda/\mu)^j e^{-\lambda/\mu} / j!}{\sum_{i=0}^K (\lambda/\mu)^i e^{-\lambda/\mu} / i!} = \frac{(\lambda/\mu)^j / j!}{\sum_{i=0}^K (\lambda/\mu)^i / i!}$$

— Erlang's loss distribution in one line.

$M/M/\infty$ and $M/M/K/K$ queues

Truncation is not only for intervals

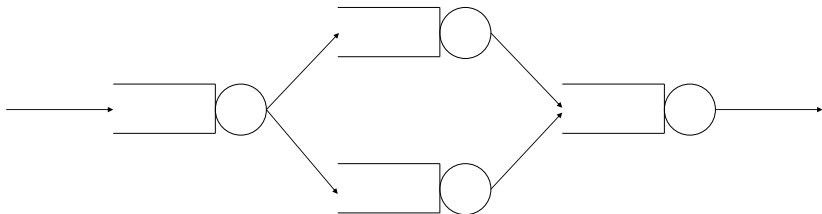


Any subset $\mathcal{S}'$ of the state space of a reversible chain works. Example: two *independent* $M/M/\infty$ queues, state (x_1, x_2) with product weights $p_1(x_1)p_2(x_2)$ (a product of reversible chains is reversible), restricted to $x_1 + x_2 \leq K$ (two call types sharing K lines) or to the staircase on the left:

$$p^B(x_1, x_2) = \frac{p_1(x_1) p_2(x_2)}{\sum_{\mathcal{S}'} p_1(y_1) p_2(y_2)}, \quad (x_1, x_2) \in \mathcal{S}'.$$

▷ The workhorse behind loss networks and many blocking models — without a single balance equation.

Network of queues



- Network types

- Open: external arrivals, departures
- Closed: conservation of customers

▷ Companion §1.6: the state of a network is the vector $X(t) = (X_1(t), \dots, X_N(t))$.

Jackson network

- Single-class open queueing network
- External arrivals are Poisson.
- Service times are exponential.
- Any number of servers in a station, including ∞ .

▷ Companion Proof 8: traffic equations, product form, tandem and feedback queues, Braess' paradox.

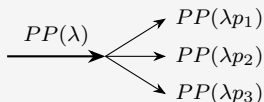
Jackson network

Two more standing assumptions

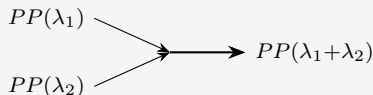
Capacity of each station is infinite (nobody is blocked or lost inside the network); all stations are stable. Routing is Markovian: after service at i go to j w.p. p_{ij} or leave w.p. $r_i = 1 - \sum_j p_{ij}$, independently of everything else.

The three tools that make the analysis work

1) Poisson splitting

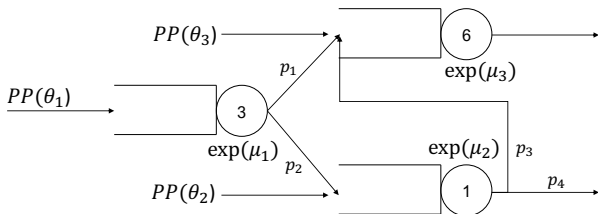


2) Merging



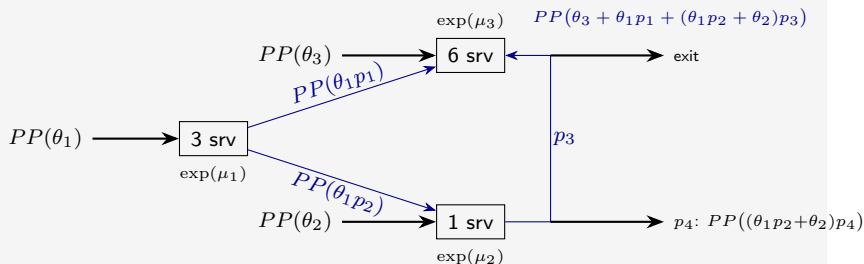
3) Time reversibility (Burke): the departure process of each $M/M/s$ station is PP in steady state — every station receives a superposition of thinned Poisson flows. (Caveat: with feedback the internal flows are not Poisson, yet the product form holds — CP13 §4.)

Example



Example

Effective arrival rates: follow the Poisson flows through the picture



Node 2 receives a split flow merged with an external flow, $PP(\theta_1 p_2 + \theta_2)$; its departures are Poisson (Burke) and split again.

Effective arrival rates:

$$\lambda_1 = \theta_1, \quad \lambda_2 = \theta_1 p_2 + \theta_2, \quad \lambda_3 = \theta_3 + \theta_1 p_1 + (\theta_1 p_2 + \theta_2) p_3.$$

Wanted: $\lim_t P(X_1(t) = i, X_2(t) = j, X_3(t) = k)$ — a CTMC on $\mathbb{Z}_+^3$ that is never solved directly.

Setting

- N nodes (stations)
- Each node i has $PP(\theta_i)$ as external arrivals. If no external arrival, $\theta_i = 0$.
- There are s_i servers at node i , $s_i \in \{1, 2, \dots\} \cup \{\infty\}$.
- Each node has infinite capacity.
- Service time at node i is $\exp(\mu_i)$.
- When a service is completed at node i , the customer joins node j with prob. p_{ij} and exit with prob. r_i .

$$\sum_{j=1}^N p_{ij} + r_i = 1.$$

▷ Companion Proof 8(a): the routing matrix $P = [p_{ij}]$ is substochastic — $I - P$ is invertible when every customer eventually leaves.

Analysis

- Assumptions
 - All stations are stable.
 - At least one node has external arrivals.
 - At least one node has departures.
- For node i ,

$$\lambda_i = \theta_i + \sum_{j=1}^N \lambda_j p_{ji}.$$

λ_i is called *effective arrival rate* to node i .

- Matrix form

$$\boldsymbol{\lambda} = \boldsymbol{\theta} + \boldsymbol{\lambda}P,$$

where $\boldsymbol{\lambda} = (\lambda_1, \dots, \lambda_N)$, $\boldsymbol{\theta} = (\theta_1, \dots, \theta_N)$, and $P = [p_{ij}]$.

$$\boldsymbol{\lambda} = \boldsymbol{\theta}(I - P)^{-1}.$$

Analysis

- Node i can be modeled as an $M/M/s_i$ queue with $PP(\lambda_i)$ arrivals and $\exp(\mu_i)$ service.
- In steady state, the number of customers at node i , $X_i(t)$, is independent of the number of customers at any other nodes.

$$\begin{aligned} & \lim_{t \rightarrow \infty} P(X_1(t) = x_1, \dots, X_N(t) = x_N) \\ &= \lim_{t \rightarrow \infty} P(X_1(t) = x_1) \cdots P(X_N(t) = x_N). \end{aligned}$$

▷ Companion Proof 8(b): the product form, proved by substituting the guess $p(\mathbf{x}) = C \prod_i \phi_i(x_i)$ into the global balance equations (the Independence handout).

► Colab notebook: CP13 §5 (split vs. pooled servers) [INCLASS] [full]

Analysis

Using the product form: node by node, then the whole network

Traffic equations $\lambda(I - P) = \theta$ first ($I - P$ is invertible because every customer eventually leaves). Then **node** i = an $M/M/s_i$ queue with arrival rate λ_i (the effective rate, *not* θ_i): $p_i(x)$, L_i from slides 17–22;

$$L_i = \frac{\lambda_i}{\mu_i - \lambda_i} \quad (s_i = 1), \quad L_i = \frac{\lambda_i}{\mu_i} \quad (s_i = \infty), \quad W_i = \frac{L_i}{\lambda_i}$$

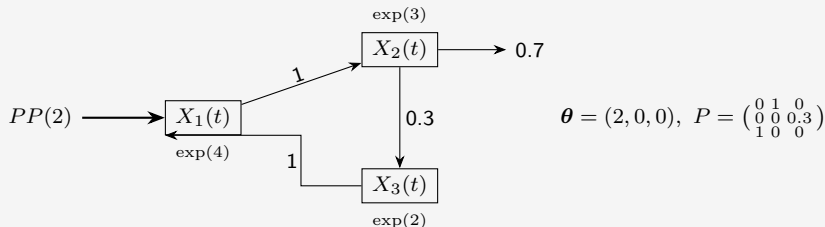
(Little at node i : W_i is the time per *visit*). **Whole network** (Little for a box around everything, arrival rate $\sum_i \theta_i$):

$$L = \sum_{i=1}^N L_i, \quad W = \frac{L}{\sum_{i=1}^N \theta_i} \quad \left(= \sum_i v_i W_i, \quad v_i = \frac{\lambda_i}{\sum_k \theta_k} \right).$$

▷ The X_i are independent in steady state even where the flow into node i is not Poisson; only the stationary law factorises, not the process (Companion Proof 8).

Example

A three-station network with feedback: $PP(2)$ into station 1, service rates 4, 3, 2



Traffic equations: $\lambda_1 = 2 + \lambda_3$, $\lambda_2 = \lambda_1$, $\lambda_3 = 0.3\lambda_2 \Rightarrow \lambda_1 = 2/0.7$:

$$\lambda = \theta(I - P)^{-1} = (2.8571, 2.8571, 0.8571).$$

▷ All three $\rho_i = \lambda_i/\mu_i$ (0.714, 0.952, 0.429) are below 1: stable, with station 2 as the bottleneck.

► Colab notebook: CP13 §1 (three-station network) [INCLASS] [full]

Example

Each node is $M/M/1$ with its own λ_i

$$L_1 = \frac{2.8571}{4 - 2.8571} = 2.5, \quad L_2 = \frac{2.8571}{3 - 2.8571} = 20,$$

$$L_3 = \frac{0.8571}{2 - 0.8571} = 0.75; \quad W_i = \frac{L_i}{\lambda_i} : W_1 = 0.875, W_2 = 7, W_3 = 0.875$$

(per visit).

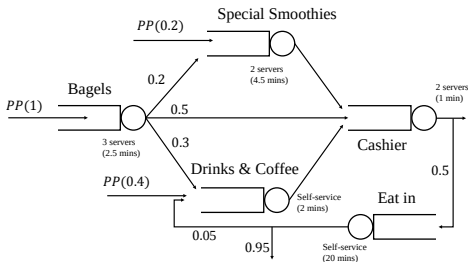
The network

$$L = L_1 + L_2 + L_3 = 23.25 \text{ customers};$$

$$W = L/\theta_1 = 23.25/2 = 11.63 \text{ time units per customer.}$$

Check via visits: a customer visits stations 1 and 2 on average $1/0.7 = 1.43$ times and station 3 0.43 times, and $1.43(0.875 + 7) + 0.43 \cdot 0.875 = 11.63 \checkmark$.

Example: Bay-Gull Bagels (handout)



Bay-Gull Bagels is a bagel store in downtown Pohang. All arrival processes are Poisson and all service times exponential (rates per minute, service times in minutes); a “self-service” station has infinitely many servers; all queues have infinite capacity. Model the store as a Jackson network and obtain the average number of customers at each of the 5 stations and in the whole store.

Example: Bay-Gull Bagels (handout)

Step 1: traffic equations

Stations: 1 Bagels ($s = 3, 1/\mu = 2.5$), 2 Smoothies ($s = 2, 4.5$), 3 Drinks ($s = \infty, 2$), 4 Cashier ($s = 2, 1$), 5 Eat in ($s = \infty, 20$);
 $\theta = (1, 0.2, 0.4, 0, 0)$ per minute.

$$\lambda_1 = 1, \quad \lambda_2 = 0.2 + 0.2\lambda_1, \quad \lambda_3 = 0.4 + 0.3\lambda_1 + 0.05\lambda_5, \\ \lambda_4 = 0.5\lambda_1 + \lambda_2 + \lambda_3, \quad \lambda_5 = 0.5\lambda_4.$$

Substituting, $\lambda_4 = 0.9 + \lambda_3$, $\lambda_5 = 0.45 + 0.5\lambda_3$, $\lambda_3 = 0.7225 + 0.025\lambda_3$,
 so

$$\lambda = (1, 0.4, 0.741, 1.641, 0.821) \text{ per minute}.$$

Step 2: stability

$$\rho_1 = \frac{1}{3 \cdot 0.4} = 0.833, \quad \rho_2 = \frac{0.4 \cdot 4.5}{2} = 0.9, \quad \rho_4 = \frac{1.641}{2} = 0.821 \text{ —}$$

all below 1; the self-service stations are always stable.

Example: Bay-Gull Bagels (handout)

Step 3: five independent $M/M/s$ queues

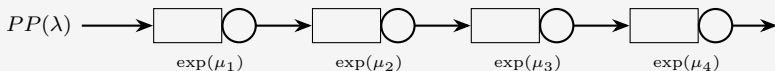
station	model	$a_i = \lambda_i / \mu_i$	ρ_i	L_i
1 Bagels	$M/M/3$	2.5	0.833	6.01
2 Smoothies	$M/M/2$	1.8	0.9	9.47
3 Drinks & Coffee	$M/M/\infty$	1.482	—	1.48
4 Cashier	$M/M/2$	1.641	0.821	5.02
5 Eat in	$M/M/\infty$	16.41	—	16.41
whole store				$L = 38.4$

$M/M/\infty$: $L_i = a_i$; $M/M/s$: $L_i = L_{q,i} + a_i$ with the L_q of slide 19.
 $W = L / (1 + 0.2 + 0.4) = 24.0$ min per customer. Smoothies ($\rho = 0.9$) holds more people than Bagels although it carries less than half the flow.

▷ CP13 §2 confirms the table by a discrete-event simulation of the store — and shows what breaks when a service time is made deterministic.

Tandem queue

Stations in series: the simplest Jackson network



$p_{i,i+1} = 1$, $r_N = 1$, $\theta = (\lambda, 0, \dots, 0)$: the traffic equations give $\lambda_i = \lambda$ at every station — what goes in must come out. By Burke each station sees Poisson(λ) input, and the product form gives

$$P(X_1 = x_1, \dots, X_N = x_N) = \prod_{i=1}^N (1 - \rho_i) \rho_i^{x_i}, \quad \rho_i = \frac{\lambda}{\mu_i} < 1.$$

► The numbers at the stations are independent at a fixed time although every customer passes through all of them — the surprise of Burke's theorem.

► Colab notebook: CP13 §3 (tandem: Burke's theorem) [INCLASS] [full]

Tandem queue

L and W

$$L = \sum_{i=1}^N \frac{\lambda}{\mu_i - \lambda}, \quad W = \frac{L}{\lambda} = \sum_{i=1}^N \frac{1}{\mu_i - \lambda}$$

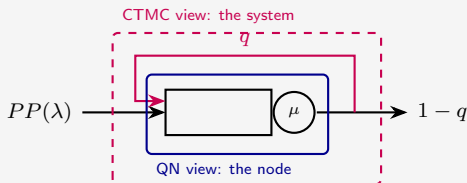
— the sojourn is the sum of the N station sojourns, each $\exp(\mu_i - \lambda)$ as in an isolated $M/M/1$ (for single servers they are even independent: Reich's theorem).

Example

$\lambda = 2, \mu = (4, 3)$: $L = \frac{2}{4-2} + \frac{2}{3-2} = 1 + 2 = 3$, $W = 1.5$ — the slower second station carries two thirds of the congestion.

Feedback queue

One station, two viewpoints



After each service the customer leaves w.p. $1 - q$ or rejoins the queue w.p. q .

1) CTMC. The number in the system is a birth–death chain with birth rate λ and death rate $\mu(1 - q)$ (a fed-back service does not change the state): an $M/M/1$ with service rate $\mu(1 - q)$,

$$L = \frac{\lambda}{\mu(1 - q) - \lambda}, \quad W = \frac{1}{\mu(1 - q) - \lambda}.$$

Feedback queue

2) Queueing network view

One node with $p_{11} = q$: $\hat{\lambda} = \lambda + q\hat{\lambda}$, so $\hat{\lambda} = \lambda/(1 - q)$ (external + fed-back customers); the node is $M/M/1$ with rates $\hat{\lambda}$, μ :

$$L = \frac{\hat{\lambda}}{\mu - \hat{\lambda}} = \frac{\lambda/(1 - q)}{\mu - \lambda/(1 - q)} = \frac{\lambda}{\mu(1 - q) - \lambda} \quad \checkmark,$$

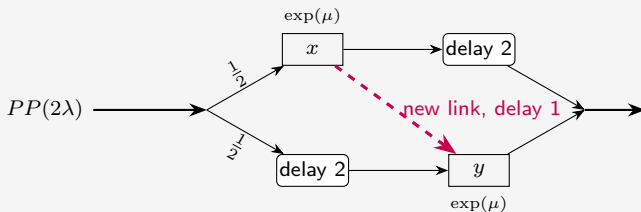
$$W_{\text{node}} = \frac{L}{\hat{\lambda}} = \frac{\lambda}{\mu(1 - q) - \lambda} \cdot \frac{1 - q}{\lambda} = \frac{1 - q}{\mu(1 - q) - \lambda}.$$

Same L , two different W 's

The QN W is per *visit* (one pass through the node), the CTMC W is per *customer*: a customer makes $1/(1 - q)$ visits on average, and $W_{\text{customer}} = W_{\text{node}}/(1 - q)$. Always ask which arrival rate Little's law is applied with. (The flow into the node, external + feedback, is not Poisson — the product form does not need it to be.)

Braess' paradox

Two parallel routes, each a queue plus a fixed delay of 2



Before the link. Half of the traffic takes each route: each queue is $M/M/1$ with rate λ (Poisson splitting), sojourn $1/(\mu - \lambda)$,

$$W_x = W_y = \frac{1}{\mu - \lambda} + 2 \quad \Rightarrow \quad W = \frac{1}{\mu - \lambda} + 2.$$

▷ x and y also denote the mean sojourn times at the two queues; each depends on the flow it receives, $x = 1/(\mu - f_x)$.

Braess' paradox

After the link: customers choose the fastest route for themselves

Three routes: $x \rightarrow 2$, $x \rightarrow 1 \rightarrow y$, $2 \rightarrow y$. In equilibrium every used route takes the same time (nobody can gain by switching):

$$x + 2 = x + 1 + y = 2 + y \implies x = y = 1 \implies W = 3.$$

($x = 1$ means $1/(\mu - f_x) = 1$: each queue carries flow $\mu - 1$; all three routes are used when $\lambda + 1 < \mu \leq 2\lambda + 1$.)

The paradox

If $\mu > \lambda + 1$: $W_{\text{before}} = \frac{1}{\mu - \lambda} + 2 < 3 = W_{\text{after}}$ — adding a free, fast link made *everyone* slower. $\lambda = 1, \mu = 2.5$: $2.67 \rightarrow 3$.

▷ Selfish routing is not socially optimal; a central planner would ignore the new link. CP13 §6 computes the flows and simulates both networks.